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Analysis of Trusses

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Analysis of Trusses

- A truss is a structure that consists of **two-force members only**, where members are organized so that the assemblage as a whole behaves as a single object.

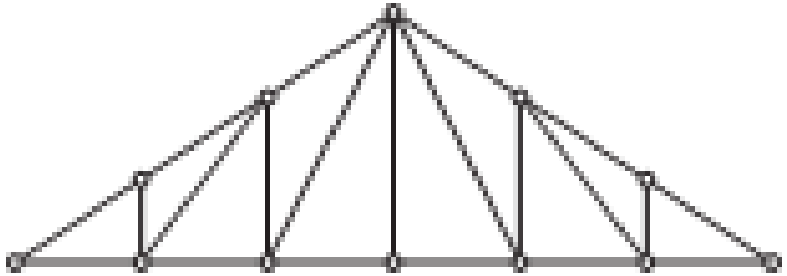
- For a two-dimensional structure to be a plane truss, it must have the following characteristics:

  ✓ All members must be connected to one another by frictionless **pins**. The locations of these pins are called **joints**.

  ✓ Each member may have no more than two joints.

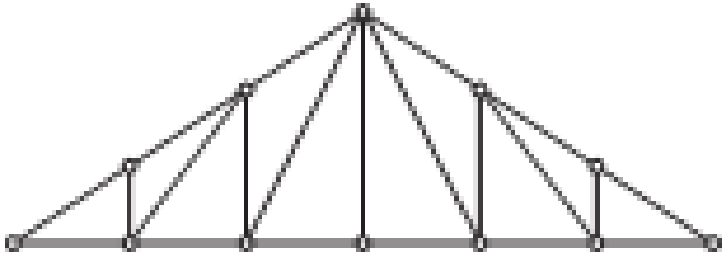
  ✓ Forces may be applied at joints only.

  ✓ The weight of individual members must be negligible.



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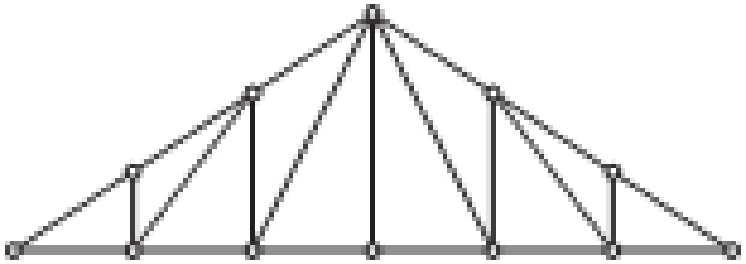
- The objectives in analysis of a truss are:
  - ✓ Determination of the support reactions for the truss, and
  - ✓ Determination of the forces supported by the individual members of the truss.



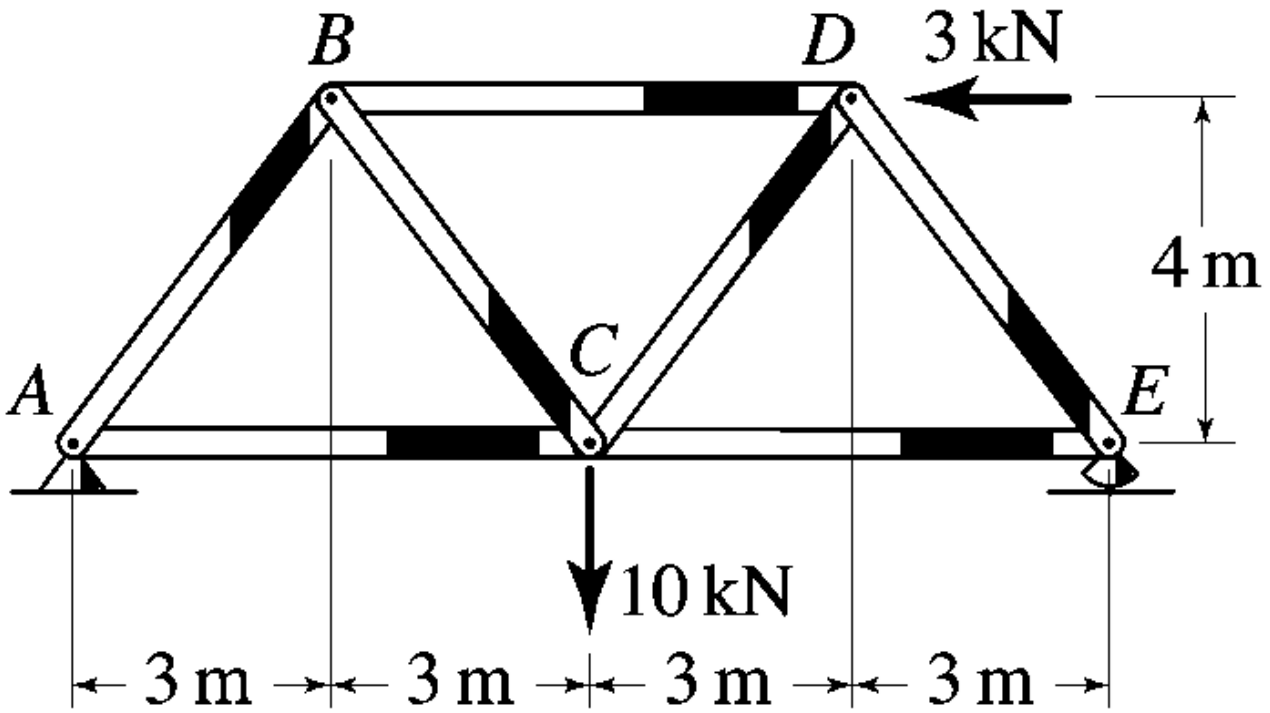
- In the *method of joints*, a truss is analyzed by treating each joint as a particle.
- If the entire truss is in equilibrium, then every joint within the truss is also in equilibrium.

Analysis of Trusses

- The analysis proceeds by drawing FBDs of the joints through-out the truss,
- Then writing equilibrium equations for each of the joints, and solving the equilibrium equations for the unknowns.
- Prior to this, it may be necessary or desirable to determine the reactions for the truss as a whole.



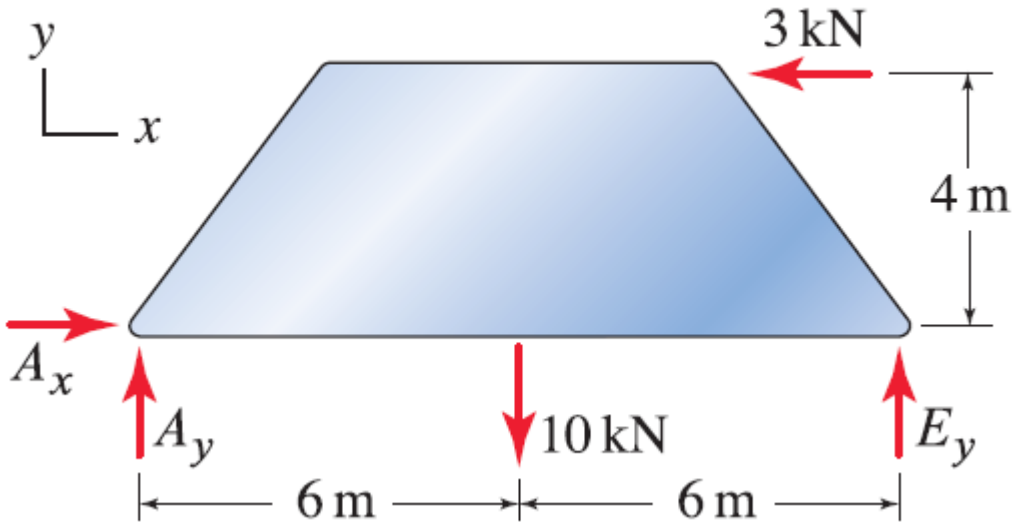
Analysis of Trusses



Method of joints:

Use the method of joints to determine the force supported by each member of the truss shown.

Analysis of Trusses



Method of joints:

Obtain the support reactions for the truss by drawing the FBD of the structure as a whole.

Next, equilibrium equations are written:

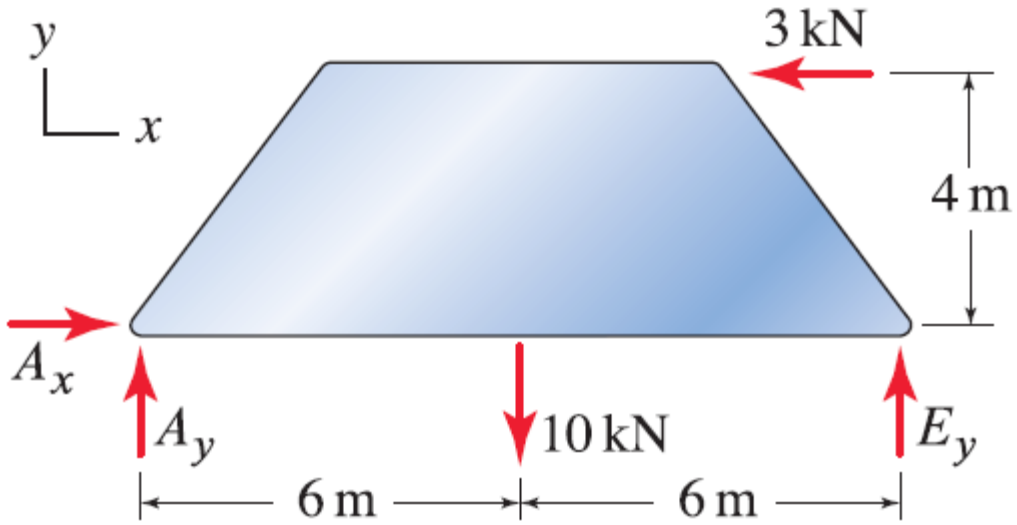
$$\sum M_A = 0 \qquad -(10\,\text{kN})(6\,\text{m}) + E_y(12\,\text{m}) + (3\,\text{kN})(4\,\text{m}) = 0$$

$$E_y = 4\,\text{kN}$$

$$\sum F_y = 0: \qquad A_y + E_y - 10\,\text{kN} = 0$$

$$A_y = 6\ kN$$

Analysis of Trusses

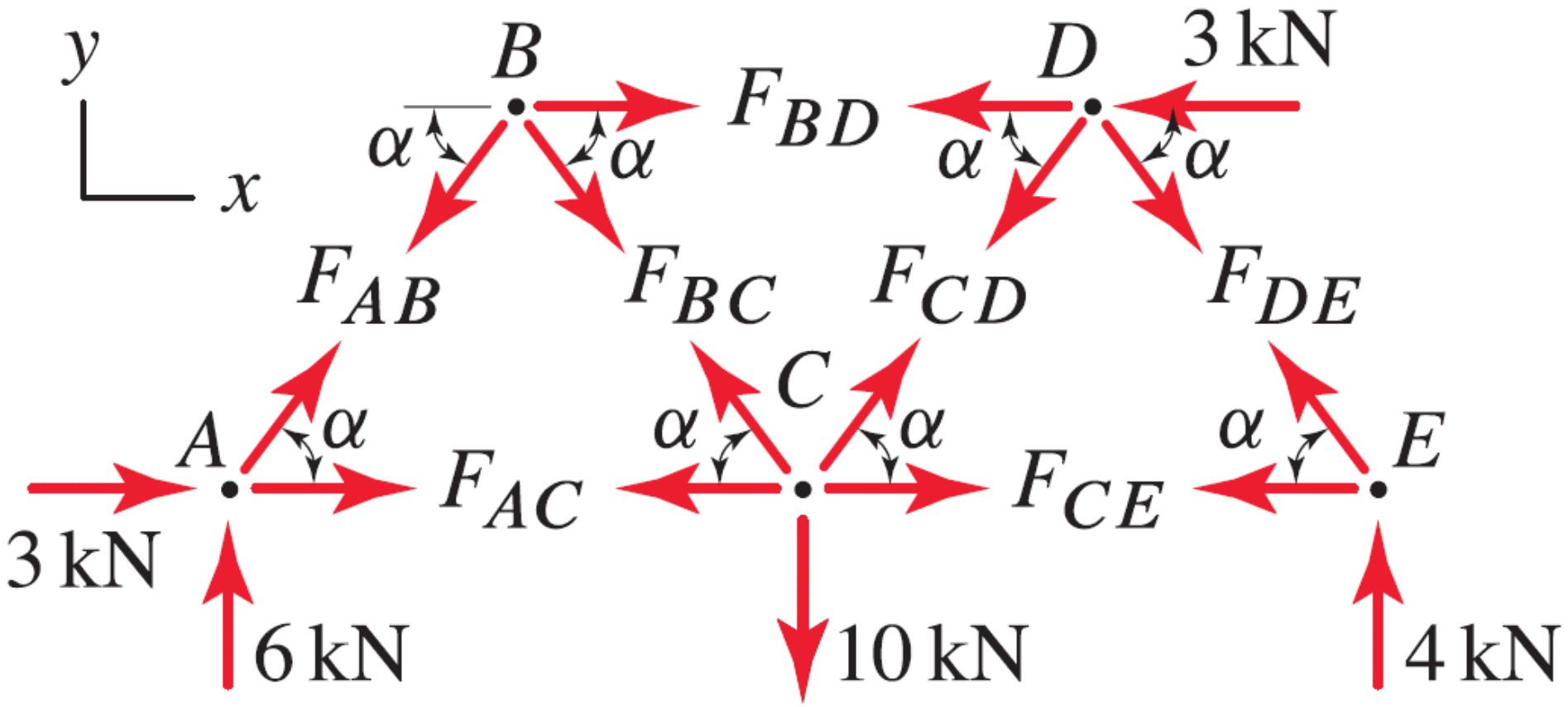


$$\sum F_x = 0: \quad A_x - 3\,\text{kN} = 0$$

$$A_x = 3\,\text{kN}.$$

Next step: Free-Body Diagrams are drawn for each joint (pin)

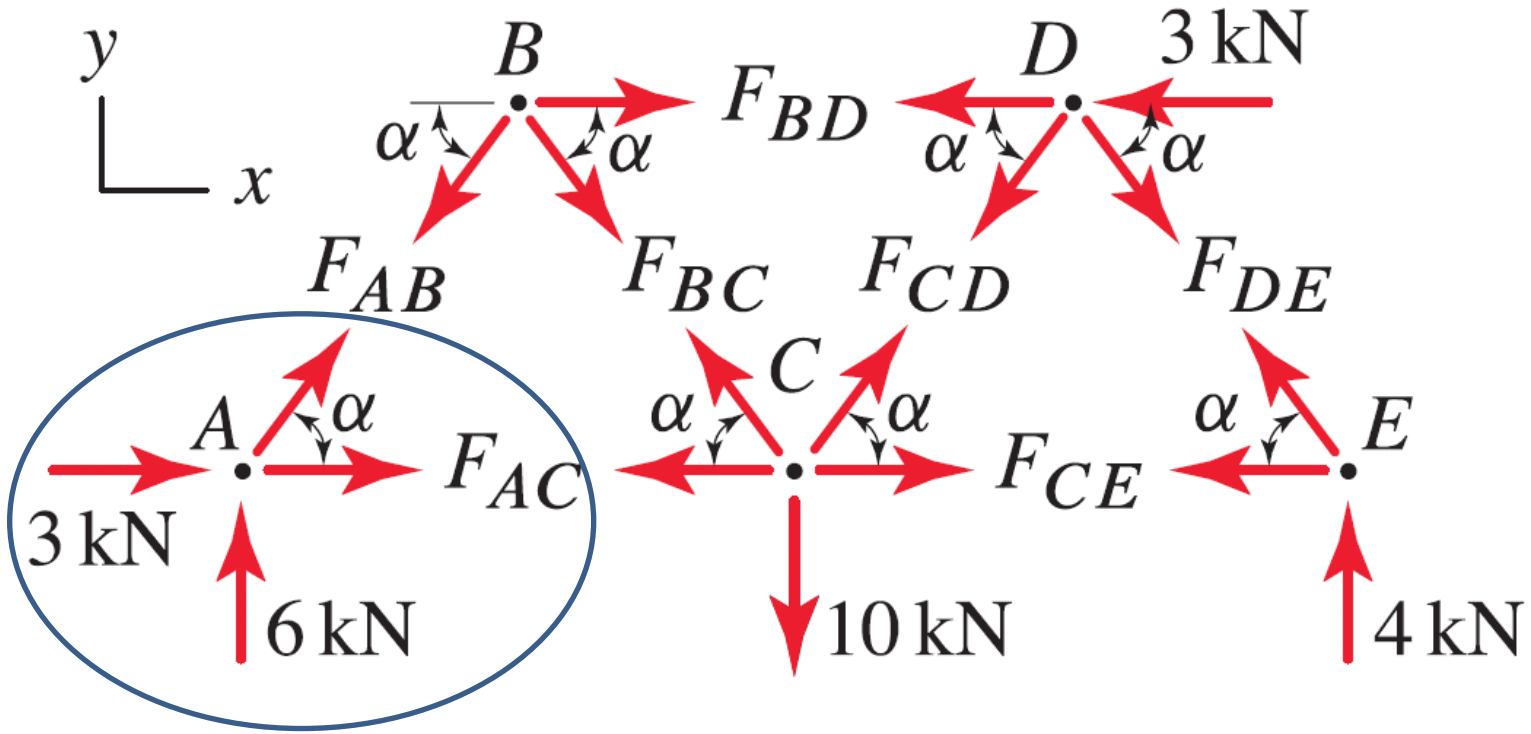
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$$\alpha = \tan^{-1}\left(\frac{4}{3}\right) = 53.1^{o}$$

Each joint has a concurrent force system and is therefore treated as a particle in equilibrium.

Analysis of Trusses

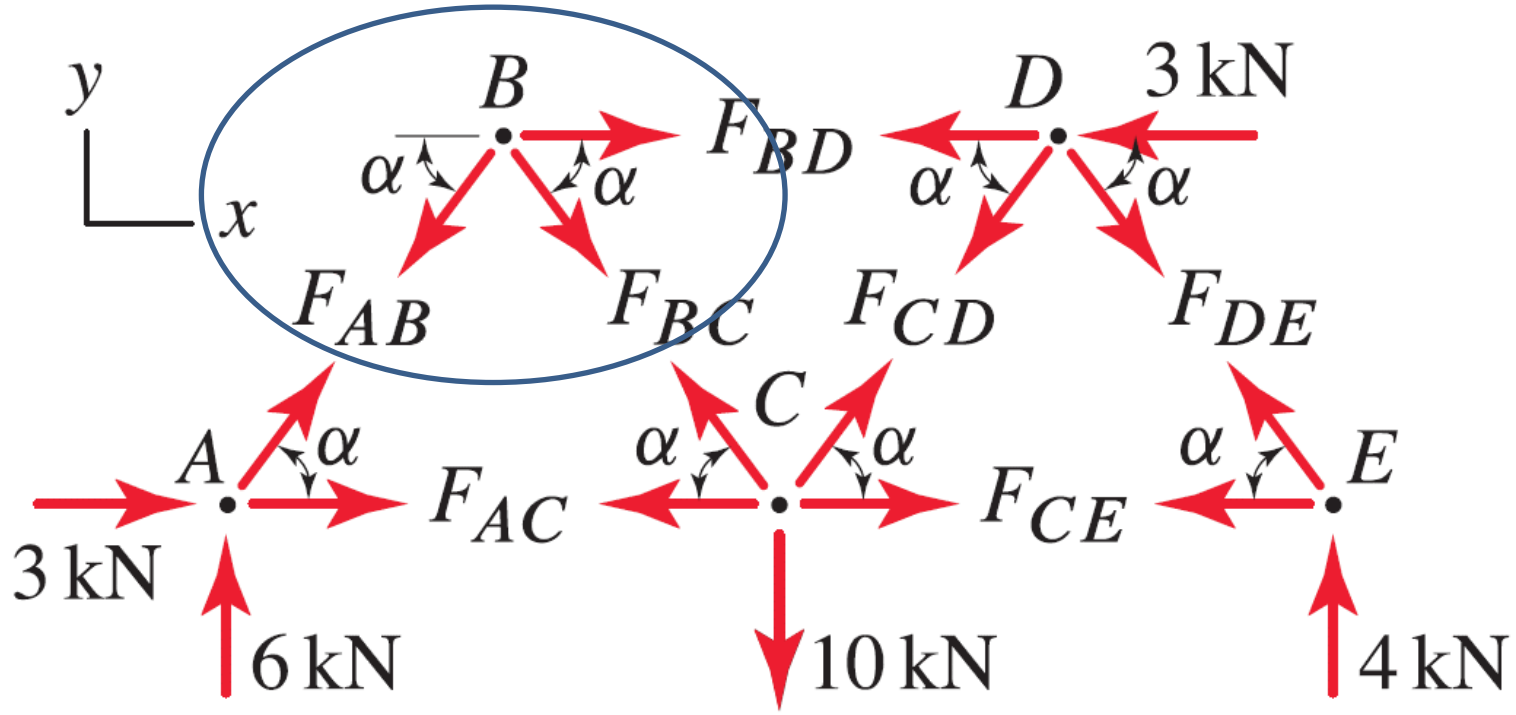


Joint A

$$\sum F_y = 0: \quad 6\,\text{kN} + F_{AB}\sin\alpha = 0 \quad \Rightarrow \quad F_{AB} = -7.5\,\text{kN}$$

$$\sum F_x = 0: \quad 3\,\text{kN} + F_{AB}\cos\alpha + F_{AC} = 0 \quad \Rightarrow \quad F_{AC} = 1.5\,\text{kN}$$

Analysis of Trusses

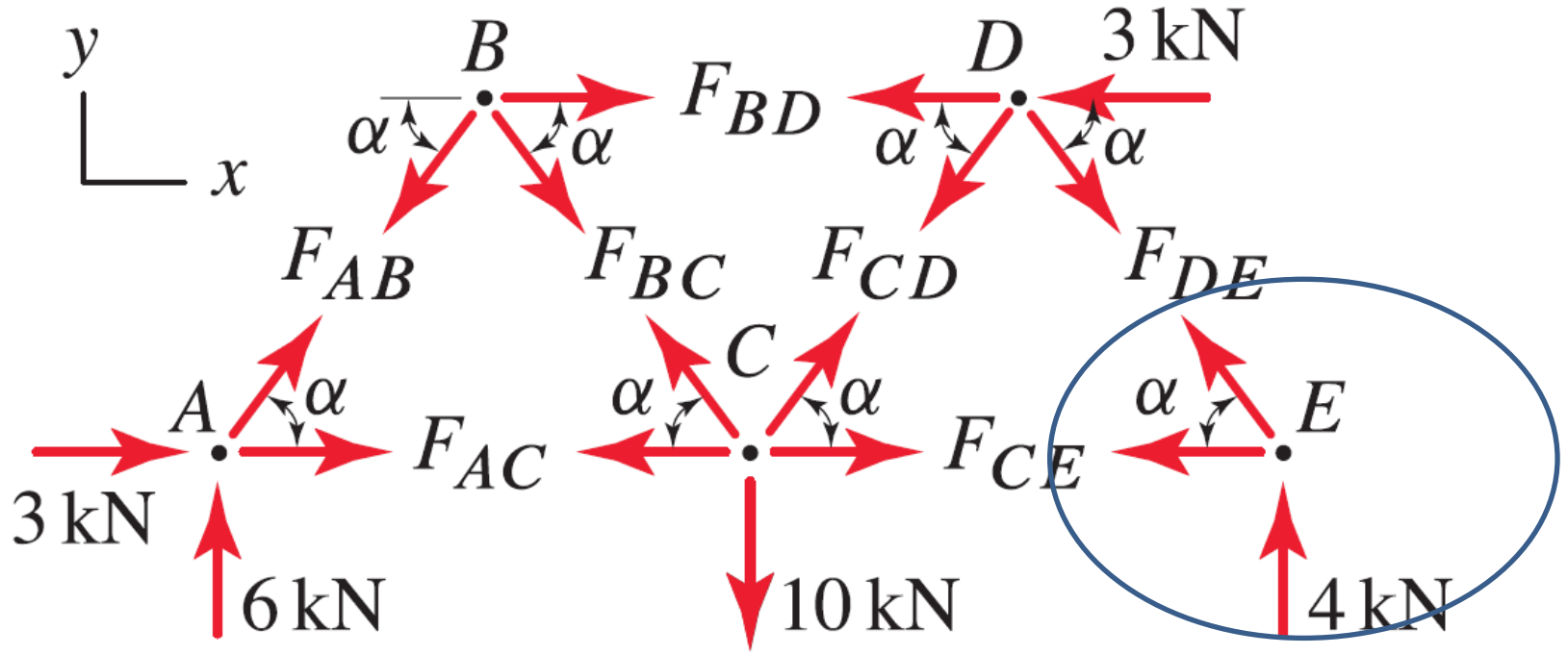


Joint B

$$\sum F_y = 0: \quad -F_{AB}\sin\alpha - F_{BC}\sin\alpha = 0 \quad \Rightarrow \quad F_{BC} = 7.5\,\text{kN}$$

$$\sum F_x = 0: \quad -F_{AB}\cos\alpha + F_{BC}\cos\alpha + F_{BD} = 0 \quad F_{BD} = -9.0\,\text{kN}.$$

Analysis of Trusses

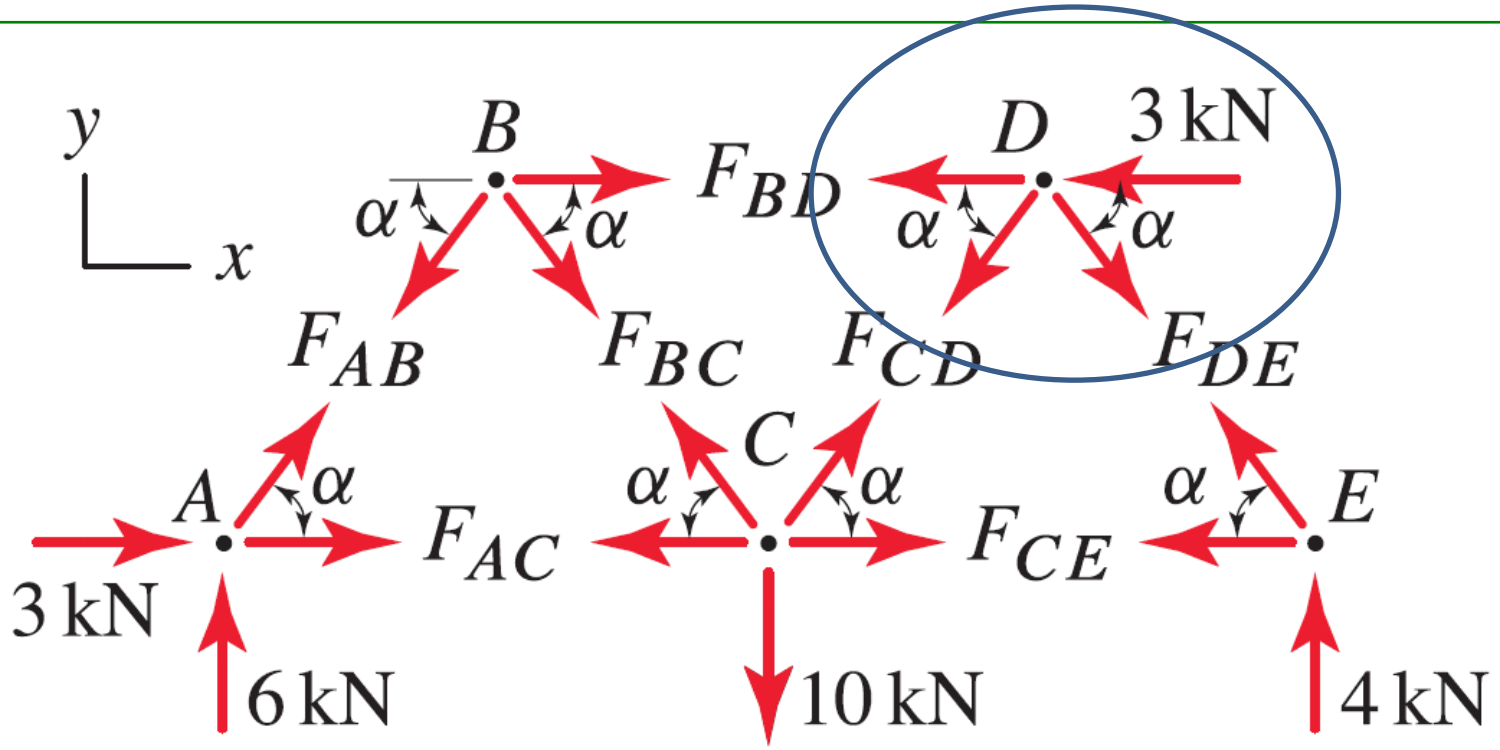


Joint E

$$\sum F_y = 0: \quad 4\,\text{kN} + F_{DE}\sin\alpha = 0 \quad \Rightarrow \quad F_{DE} = -5\,\text{kN}$$

$$\sum F_x = 0: \quad -F_{CE} - F_{DE}\cos\alpha = 0 \quad \Rightarrow \quad F_{CE} = 3\,\text{kN}$$

Analysis of Trusses



Joint D

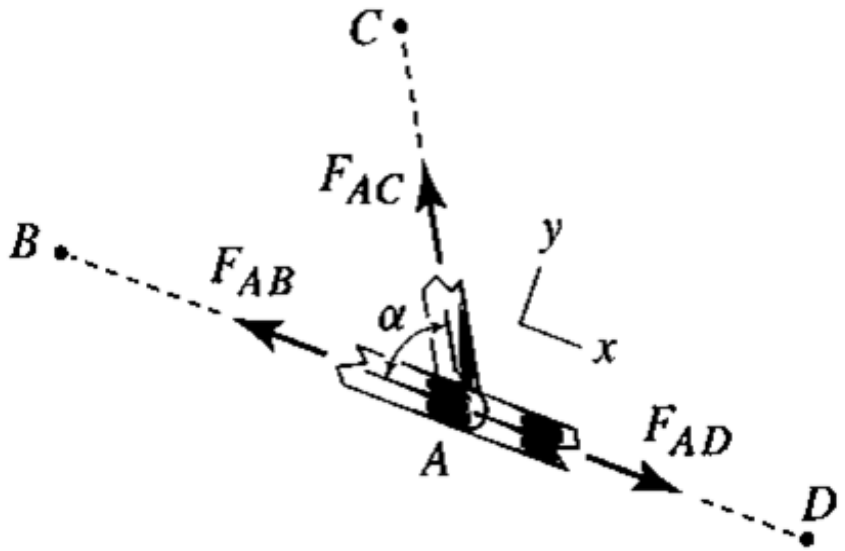
$$\sum F_y = 0: \quad -F_{CD}\sin\alpha - F_{DE}\sin\alpha = 0$$

$$\Rightarrow \quad F_{CD} = 5\,\text{kN}$$

Zero-force members

- A truss member (or any member) that supports no force is called a zero-force member.

- Joint **A** connects three truss members, two of the members are collinear (**AB** and **AD**), and no external force is applied to joint A.



- With the ***xy*** coordinate system shown, summing forces in the y direction provides:

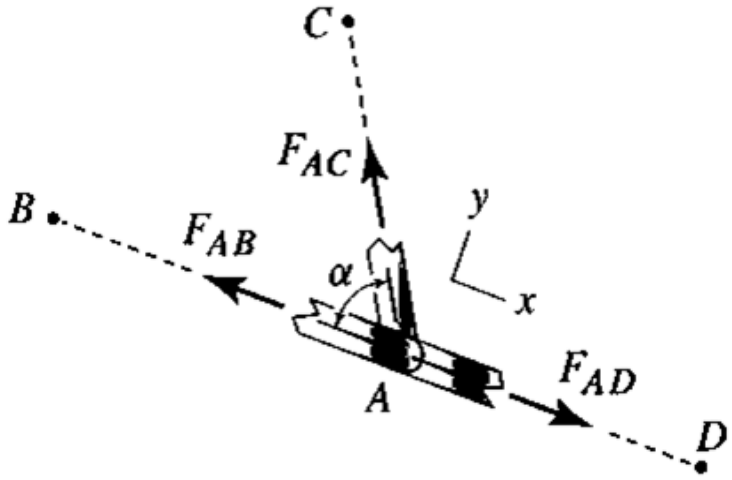
$$\sum F_y = 0: \quad F_{AC} \sin\alpha = 0$$

Except for α = 0° or 180°, F_{AC} is 0 and AC is a zero-force member of the truss.

F_{AC} will not be zero if an external force were acting on joint A.

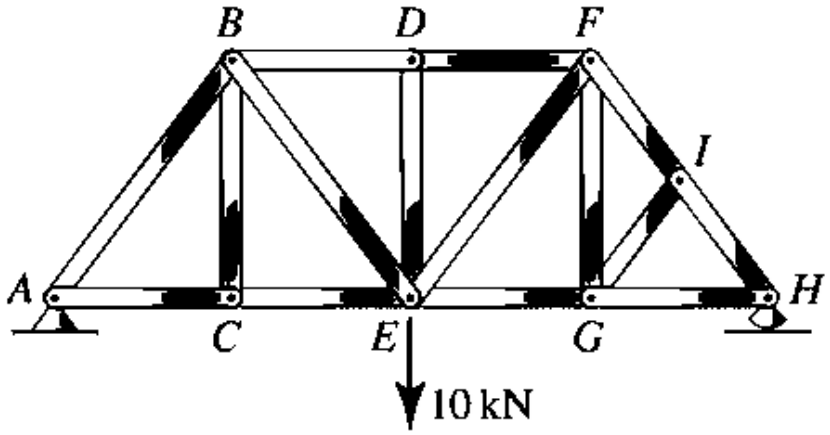
Zero-force members

- The following steps may be used to identify a zero-force member in a truss.
  - ✓ The joint must have three members connected to it.
  - ✓ Two of the members are collinear.
  - ✓ The joint has no external force applied to it.
  - ✓ Then the noncollinear member is a zero-force member.



Zero-force members

- Identify the zero-force members in the truss shown.



  - ✓ Examination of joint C: BC is a zero-force member.
  - ✓ Examination of joint D: DE is a zero-force member.
  - ✓ Examination of joint I: GI is zero-force member.
  - ✓ Because member GI is zero-force, examination of joint G then shows that member FG is zero-force.